

Interference of Two Particle-like States with a Spread Phase: Feature Points of Relative Configuration and a Structural Correspondence with Statistics (Observational Note, Draft)

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Character of this note

This is an observational note in the form of a thought experiment. It is not a proof paper or a claim paper.

This note is a sequel to the preceding note [1] (Central projection of a vacuum universe and particle-like states with a spread phase). The preceding note defined a single particle-like state as a phase-bearing local mode with a spread phase on the subjective geometry $S^4(R_U)$, going as far as setting its spread to $W_a = 2R_a$ (a model convention), and explicitly deferred the relative phase, interference, and statistics of multiple particles to a subsequent note. The present note treats the minimal case of that deferred part—**two particle-like states with the same spread phase**—on a closed one-dimensional relational setting.

What distinguishes this note from the preceding one is that most of its conclusions are obtained as **closed elementary Fourier-series computations**. That is, we obtain in closed form the spectrum and the total squared norm of the superposition of two particles represented as rectangular standing waves, and from these we observe the discrete feature points appearing within the continuous relative position (fusion, orthogonality, complete cancellation) and the admissibility of overlap depending on symmetrization / antisymmetrization (a structural correspondence with the exchange symmetry of quantum statistics).

This note does not modify standard quantum theory, general relativity, or special relativity. It proposes no new physical theory, proves no new mathematical theorem, and changes no observable prediction. What it does is to re-arrange and observe, from the single viewpoint of the relative configuration of two particles, known structures—the Fourier series of a square wave and Parseval's relation [4], the exchange symmetry of quantum statistics (Bose/Fermi), and the spread phase [1] and central projection [2,3] of the preceding notes.

This note does not assert any of the following:

- any change to the mathematical predictions of the Standard Model, quan-

- tum theory, or relativity;
- identification of a particle-like state with a concrete particle such as an electron, photon, or quark;
- derivation of the spin-statistics theorem, the Hilbert-space structure, or quantum field theory;
- any new observable prediction, or proof of any new mathematical theorem;
- that the interference structure of this note “derives” actual quantum statistics (we remain at the level of observing a structural correspondence).

Evaluation and interpretation are left to the reader.

Abstract

In the preceding note [1], a particle-like state was defined as a phase-bearing local mode with a spread phase on the subjective geometry, with spread $W_a = 2R_a = 2/\nu_{R,a}$ (a model convention). Here we superpose two particle-like states with the same frequency ν on a closed one-dimensional relational setting and observe their interference. Representing each particle as a rectangular standing wave consisting of odd harmonics, the amplitude of the n -th harmonic of the superposition, in terms of the relative position f measured in fundamental wavelengths, is $A_n = \frac{2}{n} \cos(n\pi f)$, and the total squared norm (an energy-like quantity) is the triangle wave $E(f) = \frac{\pi^2}{4}(1 - 2f)$ ($0 \leq f \leq \frac{1}{2}$; the autocorrelation of a square wave). From this we observe that within the continuous relative position f there appear discrete feature points corresponding to fusion ($f = 0$), orthogonality / independent coexistence ($f = \frac{1}{4}$), and complete cancellation ($f = \frac{1}{2}$). Furthermore, depending on whether the superposition is symmetrized or antisymmetrized, complete overlap ($f = 0$) is admissible (boson-like) or makes the composite field vanish, i.e. exclusive (fermion-like). Finally, for unequal frequencies, the strength of interference is governed by whether odd harmonics are shared (commensurability); without sharing, interference vanishes and the placement becomes free. All of this is presented as observations of structural correspondence; no derivation of quantum statistics is claimed.

§1 Starting point—connection to the preceding note

The preceding note [1] observed that a particle-like state with a spread phase, $P_a = (\nu_a, R_a)$, can be placed on the subjective geometry $S^4(R_U)$ produced by the central projection of an exterior-free vacuum universe. There, on the two grounds that

- position is relational, and no absolute position phase is defined for a single particle, and
- on the curved subjective geometry a simple difference of phases is not coordinate-invariant,

the formulation of the relative phase, interference, and statistics of multiple particles was deferred to a subsequent note.

This note treats the **minimal case** of that deferred part. That is, we adopt a **closed one-dimensional relational setting** in which the relative phase is straightforwardly defined (a period-1 phase line, the minimal model corresponding to a single great circle of S^4), and place on it two particle-like states with the same frequency ν . The coordinate-invariant formulation of the relative configuration on a general $S^4(R_U)$ remains, as in the preceding note, beyond the scope of this note. What this note treats is the closed computation of interference in this one-dimensional minimal setting.

Following the convention of the preceding note, the frequency ν is read as the number of times the phase makes one full turn (2π) ($\nu = 1 \leftrightarrow 360^\circ$). The relative position of the two particles is represented by the dimensionless quantity

$$f = \frac{\Delta\phi_{ab}}{2\pi} \in [0, 1),$$

the relative phase $\Delta\phi_{ab}$ measured in fundamental wavelengths (fundamental periods). $f = 0$ corresponds to coincidence, $f = 1$ to a separation of one fundamental wavelength.

§2 Representing a particle-like state as a rectangular standing wave

In the model convention of the preceding note, a particle-like state is placed not as a single sine but as a non-moving, idealized standing profile (its stability is not judged—preceding note [1]). In this note we represent this standing profile as a **rectangular standing wave**. This is due to the mathematical simplicity that it has half-wave antisymmetry (the property that shifting by half a period reverses the sign) and that its Fourier series consists of **odd harmonics only** [4]. This note assigns no physical meaning to the wave height (the absolute scale of amplitude), and concentrates on the interference structure that depends on the relative position f . The detailed shape of the profile is, as the idealization of the spread phase of the preceding note, beyond the scope of this note and is deferred to a subsequent dynamical consideration.

The Fourier series of the square wave is

$$\square(x) = \sum_{n \text{ odd}} \frac{1}{n} \sin(2\pi nx), \quad n = 1, 3, 5, \dots$$

(The normalization is taken as a sine series of amplitude $1/n$. The ordinary unit-amplitude square wave is this multiplied by the constant $4/\pi$, but the

conclusions of this note lie in the shape and zeros and do not depend on the choice of the absolute coefficient.)

The spatial profile of particle-like state a centered at position x_a (in fundamental wavelengths) is written

$$\psi_a(x) = \sum_{n \text{ odd}} \frac{1}{n} \sin(2\pi n(x - x_a)),$$

and likewise particle-like state b , centered at x_b with the same frequency,

$$\psi_b(x) = \sum_{n \text{ odd}} \frac{1}{n} \sin(2\pi n(x - x_b)), \quad x_b - x_a = f.$$

The phase line is a closed line of period 1, and x is read mod 1. By the preceding note's policy of treating ν as a formal parameter carrying a sign / phase direction, x covers one full turn including negative values.

Here the square wave $\square(x)$ is not a wave packet localized in real space, but a standing profile (a spread-phase profile) defined over the whole period line on the one-dimensional minimal model that represents the relational phase between two particles. It is read as an idealized representation, on this relational coordinate, of the local mode of the preceding note [1]. What this note treats is not localized wave packets in real space, but the superposition of rectangular standing profiles on the relational-phase coordinate.

§3 Superposition of two equal-frequency particles and its spectrum

Consider the superposition $\psi = \psi_a + \psi_b$ of the two particle-like states. Summing harmonic by harmonic and applying the product-to-sum formula

$$\sin X + \sin Y = 2 \sin \frac{X + Y}{2} \cos \frac{X - Y}{2}$$

to $X = 2\pi n(x - x_a)$, $Y = 2\pi n(x - x_b)$, with midpoint $x_c = (x_a + x_b)/2$ so that $(X + Y)/2 = 2\pi n(x - x_c)$ and $(X - Y)/2 = \pi n(x_b - x_a) = \pi n f$, we obtain

$$\boxed{\psi(x) = \sum_{n \text{ odd}} \underbrace{\frac{2}{n} \cos(n\pi f)}_{A_n} \sin(2\pi n(x - x_c))}.$$

Three points can be read off.

1. **The frequencies do not change.** The superposition still lies on the same odd-harmonic ladder $\{\nu, 3\nu, 5\nu, \dots\}$. The relative position f generates no new frequency; it merely **modulates the amplitude** of each harmonic.
2. **Position acts as a comb filter.** The amplitude of the n -th harmonic is $A_n = \frac{2}{n} \cos(n\pi f)$; f multiplies each harmonic by $\cos(n\pi f)$. Since the argument is proportional to n , **higher harmonics are filtered out faster**.
3. **Harmonic-wise extinction.** The n -th harmonic vanishes when $\cos(n\pi f) = 0$, i.e. $n\pi f = \frac{\pi}{2} + m\pi$,

$$f = \frac{2m+1}{2n} \quad (m = 0, 1, 2, \dots).$$

This is a set of discrete “partial extinction” points depending on the relative position.

§4 Squared norm (energy-like quantity) and the triangle wave

We measure the size of the superposition by the squared integral over period 1 (Parseval) [4]. Since this note introduces no dynamics, the $E(f)$ below is not a physical energy but an **energy-like quantity** based on the Parseval norm (squared norm) (no Hamiltonian or energy functional is assumed). For the same reason, the “interference” of this note is not a time-evolving phase interference but a static spatial overlap (an inner product / overlap integral). Using $\int_0^1 \sin^2(2\pi n(x - x_c)) dx = \frac{1}{2}$,

$$E(f) = \int_0^1 \psi(x)^2 dx = \sum_{n \text{ odd}} \frac{A_n^2}{2} = \sum_{n \text{ odd}} \frac{2}{n^2} \cos^2(n\pi f).$$

Using the double-angle formula and the triangle-wave identity for the odd cosine series

$$\sum_{n \text{ odd}} \frac{\cos(nx)}{n^2} = \frac{\pi}{4} \left(\frac{\pi}{2} - |x| \right) \quad (|x| \leq \pi)$$

at $x = 2\pi f$ ($0 \leq f \leq \frac{1}{2}$), together with $\sum_{\text{odd}} 1/n^2 = \pi^2/8$, we obtain

$$E(f) = \frac{\pi^2}{4} (1 - 2f) \quad (0 \leq f \leq \tfrac{1}{2}).$$

That is, this energy-like quantity is **linear in the separation f** , with

- $f = 0$ (coincidence): $E = \frac{\pi^2}{4}$ (maximum; the amplitude doubles, four times that of a single particle = fusion);
- $f = \frac{1}{2}$ (half a fundamental wavelength): $E = 0$ (**simultaneous extinction of all harmonics = complete cancellation**).

Over the whole range, $E(f)$ is a **triangle wave** of period 1 (maximal at integer f , zero at half-integer f , linear in between). Its essence amounts to the elementary fact that “**the autocorrelation of a square wave is a triangle wave.**” Indeed,

$$E(f) = \|\psi_a\|^2 + \|\psi_b\|^2 + 2\langle\psi_a, \psi_b\rangle, \quad \langle\psi_a, \psi_b\rangle = \sum_{n \text{ odd}} \frac{\cos(2\pi n f)}{2n^2} = \frac{\pi^2}{16} (1-4f),$$

so the cross-correlation term $\langle\psi_a, \psi_b\rangle$ decays linearly as a triangle wave (maximum $\pi^2/16$ at $f = 0$, minimum $-\pi^2/16$ at $f = \frac{1}{2}$). (This linear form is the branch on $0 \leq f \leq \frac{1}{2}$; for $f > \frac{1}{2}$ it continues as a period-1 triangle wave. For example, at $f = \frac{3}{4}$ it takes the triangle-wave continuation value 0, not the $-\pi^2/8$ obtained by formally substituting into the formula.)

(Note: the factor $\pi^2/8 = \sum_{\text{odd}} 1/n^2$ appears. This is itself a standard fact about the odd-index zeta sum, and the absolute coefficient depends on the amplitude normalization of §2. We do not pursue any coincidence with isomorphic numerical coefficients appearing in other contexts.)

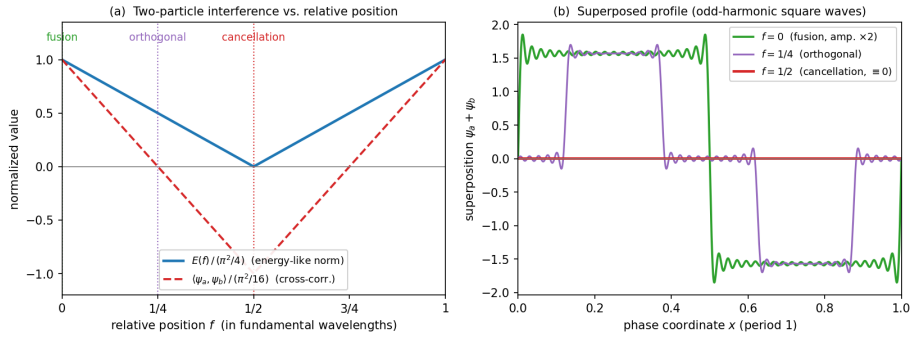


Figure 1: Figure 1

Figure 1: (a) For relative position f (in fundamental wavelengths): the squared norm $E(f)$ (triangle wave, blue) and the cross-correlation $\langle\psi_a, \psi_b\rangle$ (red dashed). Fusion at $f = 0$, orthogonality (zero cross-correlation) at $f = 1/4$, complete cancellation ($E = 0$) at $f = 1/2$. (b) The superposed profile $\psi_a + \psi_b$ (finite-term odd-harmonic square waves): doubled amplitude at $f = 0$, orthogonal at $f = 1/4$,

identically zero at $f = 1/2$. The ringing in the waveforms is the Gibbs oscillation from finite-term truncation.

§5 Feature points of the relative configuration—fusion, orthogonality, complete cancellation

The cross-correlation $\langle \psi_a, \psi_b \rangle = \frac{\pi^2}{16}(1 - 4f)$ expresses the relation of the two particles concisely. It is a linear function of the continuous variable f ; f itself is not restricted to discrete values. Within it, however, three characteristic configurations (feature points) appear.

- $f = 0$ (**fusion**): $\langle \psi_a, \psi_b \rangle$ maximal. The superposition becomes a single square wave of doubled amplitude, and the two particles become indistinguishable (effectively appearing as one).
- $f = \frac{1}{4}$ (**orthogonality**): $\langle \psi_a, \psi_b \rangle = 0$. The two square waves are orthogonal and the interference cross-term vanishes. This is a configuration in which they **can coexist independently**, the natural placement of two distinguishable particles.
- $f = \frac{1}{2}$ (**complete cancellation**): $\langle \psi_a, \psi_b \rangle$ minimal ($-\pi^2/16$). Since $\psi_b = -\psi_a$ (a square wave shifted by half a period reverses sign), $\psi \equiv 0$. In this superposition convention the composite field vanishes and no non-trivial two-particle mode remains.

Therefore, within the continuous relative position f , there appear **discrete feature points** corresponding to fusion, orthogonality, and complete cancellation. In particular, the orthogonal configurations—where the interference cross-term vanishes and the two can coexist independently—form a discrete sequence at $f = \frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \dots$ (odd $\times \frac{1}{4}$), with spacing (pitch) of half a wavelength. The orthogonal configurations sit exactly midway between the fusion points (integer f) and the complete-cancellation points (half-integer f).

To say that “the relative position is discretized,” one would have to impose a separate selection principle, e.g. restricting the admissible configurations to the orthogonal ones (independent coexistence). This note does not impose that; it keeps f a continuous variable and merely observes that the discrete sequence of orthogonal configurations appears as feature points within it. The spacing of this orthogonal sequence (half a wavelength) is not due to an external lattice, but is **the scale given by the particle’s own spread (fundamental wavelength $\propto 1/\nu_{R,a}$, preceding note [1])**. The smaller the spread (the larger $\nu_{R,a}$), the finer the spacing of the orthogonal sequence.

Furthermore, of these three feature points, the **existence** of fusion ($f = 0$), orthogonality ($f = \frac{1}{4}$), and complete cancellation ($f = \frac{1}{2}$) is **not specific to the square wave but universal to any profile with half-wave antisymmetry (odd harmonics only)**. Indeed, for a general odd-harmonic profile

$\sum_{n \text{ odd}} a_n \sin(2\pi n(x - x_a))$ the cross-correlation is $\frac{1}{2} \sum_{n \text{ odd}} a_n^2 \cos(2\pi n f)$, and at $f = \frac{1}{4}$ we have $\cos(n\pi/2) = 0$ for every odd n , so they are orthogonal independently of the amplitudes a_n . The complete cancellation at $f = \frac{1}{2}$ likewise follows immediately from half-wave sign reversal. Moreover, the modulation factor $\cos(n\pi f)$ on each harmonic (point 2 of §3) is a geometric factor arising from the offset of the two copies, independent of the amplitudes a_n ; hence the comb modulation—which harmonic vanishes where, and that higher harmonics are filtered out faster in f —is also **universal**. By contrast, only the fact that the envelope $\sum_{n \text{ odd}} a_n^2 \cos^2(n\pi f)$ formed by squaring and summing this modulation becomes **exactly piecewise linear (a triangle wave)** is specific to the square-wave (Walsh-like) choice with amplitudes $1/n$ (a pure sine gives $E \propto \cos^2(\pi f)$, which is not linear). Thus the three feature points and the comb modulation—the central message of this note—are **robust, detached from the arbitrariness of the waveform choice**, and only the linear (triangle-wave) envelope of $E(f)$ is a consequence of the square-wave choice.

This behavior **corresponds structurally** to the known structure whereby discrete levels arise from the self-consistency condition of a standing wave (quantization of states by standing waves). This note merely observes the correspondence and claims no derivation of quantization.

§6 Admissibility of overlap and a structural correspondence with quantum statistics

In the previous section we simply superposed the two particles as a sum $\psi_a + \psi_b$. When superposing identical particle-like states, there are two ways: the sum (symmetrization) or the difference (antisymmetrization). The admissibility of **complete overlap** ($f = 0$) is reversed between the two.

Symmetric combination $\psi_+ = \psi_a + \psi_b = \sum_{n \text{ odd}} \frac{2}{n} \cos(n\pi f) \sin(2\pi n(x - x_c))$

- $f = 0$: $2\psi_a$ (maximal constructive interference, doubled amplitude). At complete overlap a **non-trivial composite mode remains**. This corresponds to the behavior of stacking into the same mode.
- $f = \frac{1}{2}$: the composite field vanishes.

Antisymmetric combination $\psi_- = \psi_a - \psi_b = \sum_{n \text{ odd}} \frac{2}{n} \sin(n\pi f) \cos(2\pi n(x - x_c))$

- $f = 0$: since $\sin(0) = 0$, $\psi_- \equiv 0$. **At complete overlap the composite field vanishes, and the antisymmetric combination cannot represent a non-trivial two-particle mode.**
- $f = \frac{1}{2}$: maximal, since $\sin(n\pi/2) = \pm 1$.

That is, the amplitude-modulation factor is $\cos(n\pi f)$ for the symmetric combination and $\sin(n\pi f)$ for the antisymmetric combination, so the admissibility at $f = 0$ is exactly exchanged. In other words,

"can they completely overlap?" $\iff$ "symmetric or antisymmetric combination?"

The symmetric combination, in which complete overlap is admissible, shows the behavior of stacking into the same state (boson-like); the antisymmetric combination, which vanishes upon complete overlap, shows a behavior corresponding to the exclusion of the same configuration (Pauli-like).

This branching of overlap-admissibility by symmetrization / antisymmetrization **corresponds structurally** to the exchange symmetry of bosons / fermions in quantum statistics.

However, the $\psi_{\pm} = \psi_a \pm \psi_b$ treated here is a **one-body superposition** of two profiles on a single relational coordinate x , not the symmetrization / antisymmetrization on the two-body configuration space (x_1, x_2) , $\Psi_{\pm}(x_1, x_2) = \psi_a(x_1)\psi_b(x_2) \pm \psi_a(x_2)\psi_b(x_1)$ (symmetric Ψ_+ being bosonic, antisymmetric Ψ_- fermionic). The two are different objects. Indeed, at the coincidence point $f = 0$ where statistics genuinely act, the vanishing of the antisymmetric combination reduces to the waveform-independent algebraic identity $g - g = 0$. Hence the "structural correspondence" with quantum statistics in this note is a correspondence in the **limited sense of an isomorphism of the algebraic vanishing** ($A - A = 0$) **at the coincidence point**, not a derivation of two-body exchange symmetry.

Also, the fact that the antisymmetric combination carries a sign reversal under exchange shares a **formal commonality of sign reversal** with a mode that reverses sign upon one full turn around the closed phase (the half-integer / double-cover type). However, that the property of the permutation group S_2 and the topological property of the rotation group are non-trivially connected is itself the deep content of the spin-statistics theorem; this note does not assert that connection and confines itself to pointing out the formal commonality.

From the above, the correspondence in this note is merely an observation at the level of the superposition convention (symmetrization / antisymmetrization), and is not an explanation of the spin-statistics theorem itself, which links the symmetry of the wavefunction with spin. This note merely observes the correspondence and claims no derivation of quantum statistics.

§7 The case of unequal frequencies (observation)

So far we have treated the case where the two particles have equal frequency. We state the case of unequal frequencies briefly, as an observation.

Here we restrict to the case where the two frequencies are represented as **integer modes** on a common period line (or where a common period can be taken for a rational ratio). Sines of integer modes are orthogonal over the closed interval $(\int_0^1 \sin(2\pi m x) \sin(2\pi n x) dx = 0, m \neq n \text{ integers})$. Hence the cross-correlation of the two particles arises **only from shared harmonics**:

$$\langle \psi_a, \psi_b \rangle = \sum_{\omega \in \text{shared}} \frac{1}{2 m_\omega l_\omega} \cos(2\pi \omega f), \quad \omega = m \nu_a = l \nu_b \ (m, l \text{ odd}).$$

If no harmonic is shared, $\langle \psi_a, \psi_b \rangle = 0$ and interference vanishes regardless of the relative position f . Then the two particles can coexist independently at any relative position, and the **placement is free**.

Whether sharing occurs is determined by the integer structure of the frequency ratio. Writing ν_a/ν_b as an irreducible fraction r/s , the odd-harmonic ladders $\nu_a\{1, 3, 5, \dots\}$ and $\nu_b\{1, 3, 5, \dots\}$ share a point exactly when **r and s are both odd** (then there exist odd m, l with $ms = lr$). Normalizing to integer frequencies as $\nu_a = 2^\alpha p'$, $\nu_b = 2^\beta q'$ (p', q' odd), this corresponds to $\alpha = \beta$ (matching of the power-of-2 component). When r and s differ in parity ($\alpha \neq \beta$), no odd harmonic is shared and the result is complete orthogonality = free placement.

If ν_a/ν_b is irrational, there is no common period, and the very setting of this note—“a single closed phase line of period 1”—does not hold. The statement that interference vanishes in this case is a heuristic extrapolation outside the present setting, not a conclusion within the framework of this note.

In summary, the following correspondences are observed in the framework of this note.

- **Equal frequency (identical particles)**: all harmonics coincide, and the strong interference, sharp feature points, and statistical behavior of §3–§6 appear.
- **Unequal but commensurate (odd ratio)**: only higher harmonics are shared (with small weight $1/(ml)$), and weak interference and weak feature points remain.
- **Incommensurate (different power of 2)**: interference vanishes, free placement, independence (an irrational ratio has no common period and lies outside the present setting).

This **corresponds structurally** to the known fact that exchange statistics act only on identical particles and not on distinguishable ones. This note merely observes the correspondence.

§8 What this note does not treat

This note does not treat the following:

- the coordinate-invariant formulation of the relative configuration / interference on a general $S^4(R_U)$ (this note is restricted to the one-dimensional minimal setting);
- derivation of the spin-statistics theorem, the Hilbert-space structure, or quantum field theory;
- judgment of the stability or dynamics of a local mode (equations of motion, energy functional);
- fluctuations of particle number / uncertainty of existence, or correspondence with virtual particles;
- interference of three or more particles, the higher-dimensional structure of the configuration space;
- identification of the interference structure of this note with concrete quantum numbers, spectra, or actual statistics.

These are extension topics that could be examined if the observation of two-particle interference of this note were further developed. At the present stage, however, they go beyond the scope of this note.

§9 Scope of the claims of this note

The claims of this note are limited to the following range.

First, on a closed one-dimensional relational setting, the spectrum of the superposition of two equal-frequency rectangular standing waves stays on the same odd-harmonic ladder, with each harmonic amplitude given by $A_n = \frac{2}{n} \cos(n\pi f)$.

Second, its total squared norm (energy-like quantity) is the triangle wave $E(f) = \frac{\pi^2}{4}(1 - 2f)$ ($0 \leq f \leq \frac{1}{2}$), equal to the autocorrelation of a square wave.

Third, within the continuous relative position f there appear discrete feature points: fusion at $f = 0$, orthogonality (independent coexistence) at $f = \frac{1}{4}$, and complete cancellation (the composite field vanishes, no non-trivial mode remains) at $f = \frac{1}{2}$.

Fourth, depending on whether the superposition is symmetrized or antisymmetrized, complete overlap is admissible (boson-like) or makes the composite field vanish, i.e. exclusive (fermion-like).

Fifth, for unequal frequencies, interference arises only from shared odd harmonics, and without sharing it vanishes, giving free placement. Whether sharing occurs is determined by whether ν_a/ν_b is an odd ratio (matching of the power-of-2 component).

Sixth, all of the above are **observations of structural correspondence** with known structures (standing-wave quantization, the exchange symmetry of quantum statistics), not derivations of them.

Summary

The central observation of this note can be summarized as follows.

When two particle-like states with a spread phase are superposed as rectangular standing waves on a closed one-dimensional relational setting, discrete feature points (fusion, orthogonality, complete cancellation) appear within the continuous relative position, and the admissibility of complete overlap (boson-like / fermion-like) is divided according to symmetrization / antisymmetrization. These appear as closed elementary Fourier-series computations.

Whereas the preceding note [1] fixed the geometric stage and the spread of a single particle, this note presents, as an observation, the relational structure that appears when two particles are placed on that stage—the feature points of the relative configuration and a structural correspondence with quantum statistics. The computation itself is complete as a Fourier analysis of square waves on a period-1 closed phase line; the central projection and subjective geometry $S^4(R_U)$ of §1 are its motivating framework. The coordinate-invariant formulation on a general $S^4(R_U)$, three or more particles, and dynamics and stability are made the subject of subsequent notes.

References

- [1] N. Kihara (2026). *Central projection of a vacuum universe and particle-like states with a spread phase*. Observational note (same author; preceding note). Concept DOI: 10.5281/zenodo.20543044.
- [2] N. Kihara (2026). *A thought experiment on a model particle (continued)—the fifth frequency and composite curvature seen from the sum-to-zero condition*. Observational note (same author). Concept DOI: 10.5281/zenodo.20535894.
- [3] N. Kihara (2026). *Definition of the spherical projection σ_R (radial projection)—the foundational map of the central-projection series*. Observational note (same author). Concept DOI: 10.5281/zenodo.20462569.
- [4] T. W. Körner (1988). *Fourier Analysis*. Cambridge University Press. (Fourier series of a square wave, Parseval’s relation, autocorrelation and the triangle wave)

(Note: external references use the Concept DOI.)

Revision history

- **v1.0 (2026-06)**: Initial version (publication candidate). Treats the minimal case (two equal-frequency particles, closed one-dimensional setting)

of the multi-particle interference that the preceding note [1] deferred. Presents, all as observations of structural correspondence: the spectrum of the same-harmonic superposition $A_n = \frac{2}{n} \cos(n\pi f)$; the triangle-wave squared norm $E(f) = \frac{\pi^2}{4}(1 - 2f)$ (the autocorrelation of a square wave); the discrete feature points appearing within the continuous relative position (fusion $f=0$, orthogonality $f=\frac{1}{4}$, complete cancellation $f=\frac{1}{2}$); the reversal of overlap-admissibility by symmetrization / antisymmetrization (boson-like / fermion-like); and, for unequal frequencies, that the presence of shared odd harmonics (irreducible ratio r/s with both odd = matching power-of-2 component) governs interference. Incorporating external review (ChatGPT, Grok, Gemini, Claude.ai): terms weakened from “discretization” to “appearance of discrete feature points,” from “forbidden” to “complete cancellation / no non-trivial mode remains,” and from “emergence of statistics” to “structural correspondence with quantum statistics”; $E(f)$ stated as an energy-like quantity based on the Parseval squared norm, not physical energy; the square wave stated as a global standing profile on the relational-phase coordinate (not a localized wave packet); §7 made rigorous by restricting to integer modes / rational ratios; §2 minimally notes the functional reason for the square-wave choice (no physical meaning to wave height; shape deferred); §6 clarifies that $\psi_{\pm}$ is a one-body superposition, not the two-body antisymmetrization, the correspondence being a limited isomorphism of the algebraic vanishing $A - A = 0$ at coincidence, and weakens the double-cover relation to a “formal commonality of sign reversal”; §5 separates universal vs. waveform-specific (feature points and comb modulation universal; only the triangle-wave linear envelope specific to the $1/n$ choice). Figure 1 added in §4. No derivation of the spin-statistics theorem or quantum statistics is claimed; the general S^4 formulation, dynamics, and three or more particles are deferred to subsequent notes.